Mitchell Chatterjee

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Summary

Machine Learning Engineer II on the Agentic AI team at Layer 6 AI / TD, building long-running agents and evaluation infrastructure at enterprise scale. First author of a Scientific Reports paper on self-supervised ECG transformers.

Experience

Layer 6 AI / TD Bank Mar 2026 – Present
Machine Learning Engineer II Toronto, ON

- Developed the bank-wide standard evaluation framework for assessing agentic conversations, reusable across generative AI and other AI applications.
- Streamlined the development-to-production pipeline, reducing deployment time for custom evaluators from roughly six months to weeks.

JSI Dec 2024 – Mar 2026
AI / Machine Learning Engineer Ottawa, ON

- Led the migration of Iris, JSI's investigative AI assistant, to modular Agno/LangChain orchestration, enabling agentic retrieval, summarization, and tool use through MCP.
- Built a real-time RAG pipeline for billions of records spanning audio, text, and images, supporting on-premises deployments across customer sites with 100+ GPUs.
- Established a company-wide evaluation framework using DeepEval and A/B testing to assess retrieval relevance, latency, and interaction quality across releases.

JSI May 2019 – Sep 2022
Junior AI / Machine Learning Engineer Ottawa, ON

- Designed and deployed a real-time threat detection system that identified high-risk activity across audio, text, image, and video streams using anomaly detection and proprietary ML models.
- Introduced a Kafka-based event-driven architecture for time-sensitive alerts, reducing end-to-end notification latency by **33%**.

Research

Carleton University Sep 2022 – Aug 2024
Research Assistant Ottawa, ON

- Developed PatchECG, a 1D transformer pre-trained on 8.2 million unlabelled ECGs using masked patch modelling; achieved state-of-the-art PTB-XL multi-label classification results.

Selected Publications

PatchECG: self-supervised ECG arrhythmia detection — first author.
Scientific Reports, 2026.

AI for perioperative myocardial ischaemia: a scoping review — co-author.
British Journal of Anaesthesia, 2025.

Education

Master of Computer Science (Thesis), Carleton University 2022 – 2024
Honours BSc, Computer Science, University of Ottawa 2016 – 2020
Summa cum laude

Technical Skills

ML: PyTorch, transformers, self-supervised learning, LLM evaluation, RAG, vector search, FAISS.
Engineering: Python, SQL, Kafka, Docker, Kubernetes, LangChain, Agno, DeepEval, SLURM, Ray Tune.